PYTHON ACTIVITY-BASED LEARNING

Case Study 04: Nested Loop Calendar Generator

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Problem Statement

Generate a monthly calendar using the number of days in the month and the weekday on which the month begins.

Objectives

• Use nested loops to generate calendar rows.

• Align dates under weekday headings.

• Validate the starting weekday and month length.

Assumptions

• The month has between 28 and 31 days.

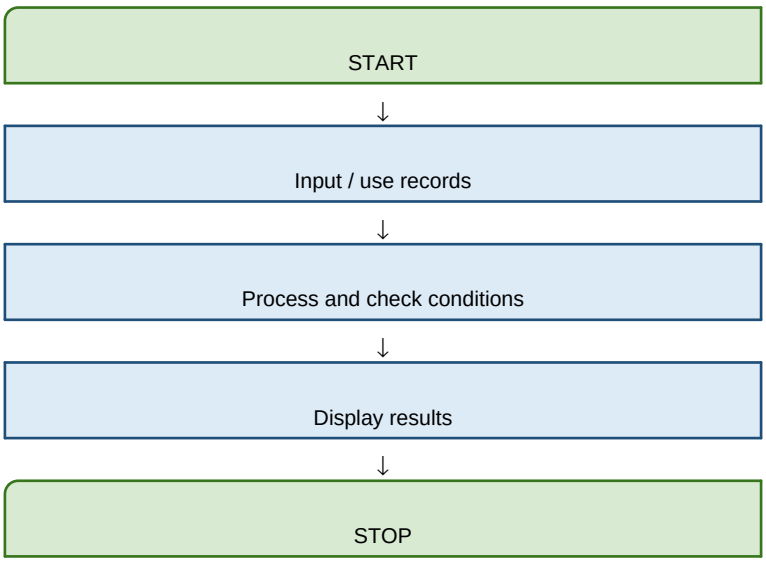
• Starting weekday is entered as 0 for Sunday through 6 for Saturday.

• The example uses 30 days and starts on Wednesday (3).

Algorithm

1. Set weekday headings and read month length and starting weekday.
2. Validate that days are between 28 and 31 and weekday is between 0 and 6.
3. Print weekday headings.
4. Print blank spaces for days before the first date.
5. Use a loop to print each date and move to a new line after Saturday.

Flowchart



Pseudocode

```
START
READ days and starting_weekday
IF values invalid, DISPLAY error and STOP
DISPLAY weekday headings
PRINT leading spaces
FOR date from 1 to days
    PRINT date
    IF (starting_weekday + date) is multiple of 7
        PRINT new line
STOP
```

Python Source Code

```
days = 30
start = 3  # 0=Sunday, 3=Wednesday

if days < 28 or days > 31 or start < 0 or start > 6:
    print("Invalid input")
else:
    print("Su Mo Tu We Th Fr Sa")
    for i in range(start):
        print("   ", end="")

    for date in range(1, days + 1):
        print("%2d " % date, end="")
        if (start + date) % 7 == 0:
            print()
    print()
```

Test Cases

Test input / condition	Expected result
Days=30, start=3	Date 1 appears under Wednesday
Days=32	Invalid input
Days=31, start=0	Date 1 appears under Sunday

Output Screenshot

Sample console output (formatted to show the expected execution result):

```
Su Mo Tu We Th Fr Sa
          1  2  3  4
 5  6  7  8  9 10 11
12 13 14 15 16 17 18
19 20 21 22 23 24 25
26 27 28 29 30
```

Conclusion

The program creates a simple aligned monthly calendar using loops and arithmetic.